PART A — Mock PAPER

SECTION A: Objective Type Questions (MCQs)

Q1. What does the success of AI primarily depend on?

A. The speed of algorithms

B. The size of the computer

C. The amount, quality, and type of data it receives

D. The location of the AI server

Q2. Which of the following is one of the three main domains of Artificial Intelligence?

A. Data Science

B. Mechanical Engineering

C. Physical Science

D. Social Science

Q3. What does Spotify primarily create based on a user's listening history?

A. Shopping ads

B. Movie suggestions

C. Job recommendations

D. Playlists

Q4. What is the main purpose of Data Science?

A. To write computer programs

B. To create computers

C. To extract meaningful insights from data

D. To design websites

Q5. How does Data Science support AI systems?

A. It acts as the backbone, helping AI systems learn and improve

B. It makes direct decisions for AI

C. It only stores data for later use

D. It creates physical hardware

Q6. What does CTR stand for in digital advertising?

A. Customer Tracking Rate

B. Click-Through Rate

C. Click Time Rate

D. Content Time Rate

Q7. Which approach allows users to build, train, and deploy AI models without needing any programming knowledge?

A. Machine Learning

B. Natural Language Processing

C. High-Code AI

D. No-Code AI

Q8. What is a key characteristic that differentiates Low-Code from No-Code development?

A. Low-Code requires manual coding by programmers

B. Low-Code is completely free

C. Low-Code requires absolutely zero coding skills

D. Low-Code is only used for hardware

Q9. What does Descriptive Statistics help identify in a dataset?

A. Complex algorithms

B. Ways to collect more data

C. Patterns, trends, and key points

D. The most complicated parts of data

Q10. In Orange Data Mining, which widget is used to load and import data into the platform?

A. Data Analyzer

B. Report Generator

C. Data Processor

D. File Widget

Q11. The complete collection of raw data available for a test or experiment is called a:

A. Variable

B. Sample

C. Population

D. Dataset

Q12. Which of the following is an open-source data mining and machine learning tool designed for data analysis, visualization, and exploration without needing coding skills?

A. Microsoft Excel

B. Orange Data Mining

C. Tableau

D. SPSS

Q13. In Probability, what represents the outcome of an experiment?

A. An event

B. A theory

C. A calculation method

D. A type of experiment

Q14. What is the large area in the center of the Orange Data Mining interface where you create workflows called?

A. Sidebar

B. Report Section

C. Canvas

D. Notification Panel

Q15. Which Evaluate Widget displays a table that compares predicted and actual class labels?

A. File Widget

B. Confusion Matrix Widget

C. Data Chart Widget

D. Report Generator Widget

Q16. Which widget is specifically used to reduce dimensionality and visualize high-dimensional data in 2D or 3D?

A. Data Reduction Widget

B. Data Visualization Widget

C. PCA Widget

D. t-SNE Widget

Q17. Which term describes the process of replacing missing values with some form of estimate?

A. Imputation

B. Data Cleaning

C. Feature Engineering

D. Data Transformation

Q18. Which of the following is a popular type of machine learning model used for making predictions?

A. Support Vector Machine

B. Random Forest

C. Decision Tree

D. Neural Network

Q19. How does standard deviation mathematically relate to variance?

A. It is the square of the variance

B. It is the square root of the variance

C. It is the variance divided by two

D. It is unrelated to variance

Q20. Teachable Machine is a web-based tool developed by which company?

A. Microsoft

B. Apple

C. Google

D. Amazon

SECTION B: Short Answer & Scenario-Based Questions

Q21. List the three main domains of Artificial Intelligence (AI).

Q22. Define Data Science and explain how it turns raw data into valuable knowledge.

Q23. A data analyst needs to prepare a dataset for advanced analysis by modifying its features. Which category of widgets should they use in Orange Data Mining?

Q24. Outline the six stages of the AI project cycle in their correct sequential order.

Q25. Explain why standard deviation is often preferred over variance when measuring how spread out values are in a dataset.

Q26. A researcher wants to study the spending habits of one million shoppers but cannot analyze every individual. Explain the concept they should use to solve this problem.

Q27. Briefly describe what Teachable Machine is and list two tasks it is especially designed to handle.

Q28. Name four popular No-Code AI tools mentioned in your study material.

Q29. Differentiate Custom Code (High Code) and No-Code development based strictly on *cost* and *flexibility*.

Q30. Provide two reasons why a marketing team with no programming background would need No-Code AI tools.

Q31. Define the "Mean" and explain the step-by-step process to calculate it.

Q32. A business uses a No-Code platform but struggles to implement a highly specific, complex feature. Identify this specific disadvantage of No-Code platforms.

Q33. Explain the "Security Issues" associated with No-Code platforms when handling private information.

Q34. Define "Automation Bias" and how it affects decision-making when using no-code tools.

Q35. How does Python integration support advanced users in Orange Data Mining?

SECTION C: Application, Distinctions & Calculations

Q36. A student received the following scores on five tests: 10, 15, 20, 25, 30. Calculate the Mean and the Median.

Q37. A small business has four employees with the following daily sales: 5, 10, 20, 25. Calculate the Median.

Q38. Describe the visual shape of a "left-skewed distribution" and state the relationship between its mean, median, and mode.

Q39. Define a "normal distribution" and explain how the mean, median, and mode relate to each other in it.

Q40. Define Probability and provide a brief explanation of how it helps us understand events.

Q41. What is the general term for the essential components in Orange Data Mining that help you interact with, manipulate, and visualize your data?

Q42. Contrast "Visualize Widgets" with "Evaluate Widgets" based on their specific functions in a machine learning workflow.

Q43. Differentiate the purpose of "Model widgets" from "Unsupervised widgets".

Q44. A local clinic decides to use Microsoft's Azure Machine Learning. Classify this tool under the appropriate development approach (High, Low, or No Code).

Q45. A global bank requires a custom security application built from the ground up using Java. Which development approach (High, Low, or No Code) must they use and why?

SECTION D: Long Answer & Multi-Part Questions

Q46. A retail store wants to predict future sales revenue using an AI model.

1. Map out the first four stages of the AI Project Cycle they must follow.
2. If they use Orange Data Mining, identify the widget category they will use to create and train this predictive model.

Q47. A climate scientist is analyzing 50 years of raw temperature data (the complete

collection).

1. Identify the statistical term for this complete collection.
2. They notice several missing temperature readings from 1980. Identify the process used to estimate and replace these values.
3. To visualize this high-dimensional dataset in 2D, name the specific widget they should apply.

Q48. A healthcare startup wants to deploy a No-Code AI system to diagnose patient illnesses automatically.

Evaluate this decision by analyzing two major disadvantages of No-Code tools: Security Issues and Automation Bias.

Q49. A financial analyst creates a chart of user incomes and notices it forms a left-skewed distribution.

1. Explain where the long tail is located on this chart.
2. In this specific distribution, will the mean be higher or lower than the median?
3. What is the primary purpose of using Descriptive Statistics on this income data instead of just looking at raw numbers?

Q50. An AI student is building a workflow in Orange Data Mining from scratch.

Describe the sequence of actions they would take on the **Canvas**, utilizing the **File Widget**, **Transform Widgets**, **Model Widgets**, and finally the **Confusion Matrix Widget**. Explain the distinct role of each widget in this workflow.